



UNIVERSITÀ DEGLI STUDI DI PISA

CORSO DI STUDIO IN BIONICS ENGINEERING

ADVANCED SIGNAL PROCESSING - *Prova scritta preliminare del 30/06/2016*

EXERCISE

We have a random variable X having uniform distribution between 0 and θ , where $\theta > 0$.

Assuming that we observe N independent realizations of X :

- (1) derive and plot the likelihood function $L(\theta)$;
- (2) derive the maximum likelihood (ML) estimator of parameter θ , its bias and mean square error (MSE), and establish if the estimator is consistent;
- (3) derive the method of moments (MM) estimator of parameter θ , its bias and mean square error (MSE), and establish if the estimator is consistent;
- (4) plot and compare the MSE's of the two estimators;
- (5) comment on the pros and cons of the two estimators.